CSCE Québec 2026
SCGC Québec 2026

Quebec City, QC
June 3-5, 2026 / 3-5 juin 2026

Physics-Informed Deep Learning with Second-Order Macroscopic Model for Robust Traffic State Estimation

Waheed Imran[1] and Achie J. Huang[2]*

[1] Department of Civil, Environmental and Architectural Engineering, University of Naples, Federico II, Via Claudio 21, 80125, Naples, Italy (waheedemran@hotmail.com)
[2] Department of Building, Civil and Environmental Engineering, Gina Cody School of Engineering and Computer Science, Concordia University, Montréal, QC H3G 1M8, Canada (archie.huang@concordia.ca)

ABSTRACT: Traffic state estimation (TSE) aims to reconstruct how traffic states evolve over space and time in terms of speed, flow, and density. Reliable TSE supports highway operations such as incident monitoring and control actions, but in practice the available data are sparse, noisy, and heterogeneous. Fixed detectors provide measurements at a few locations, while probe vehicles provide measurements along a limited number of trajectories. Further, speed is the only reliable input in many applications, whereas density is missing (e.g., TomTom services) or highly uncertain. Classical model-based estimators depend on simplified assumptions and fixed parameters and can lose accuracy when traffic deviates from the assumed dynamics. Pure data-driven deep-learning (DL) approaches often require large datasets and may produce unrealistic reconstructions when data coverage is limited. This paper proposes a velocity-only physics-informed deep learning (PIDL) model for TSE. The approach uses neural networks to represent the space-time fields of speed. It fits the available speed observations while also enforcing traffic-model consistency at additional points across the space-time domain. This combination helps the estimator remain stable and physically plausible in regions where no sensors are available. The PIDL model is evaluated on a METANET-generated benchmark under two sensing settings such as Eulerian fixed detectors and Lagrangian probe trajectories. In both cases, the proposed PIDL model improves reconstruction accuracy over a DL baseline, reducing root mean square error (RMSE) from 3.35 to 1.98 and from 2.14 to 1.62, respectively. These results suggest that velocity-only PIDL can provide more reliable speed reconstructions from realistic highway data sources, supporting practical monitoring and control applications.

1. INTRODUCTION AND BACKGROUND

Traffic state estimation (TSE) aims to reconstruct the evolution of macroscopic traffic variables, primarily density, flow, and speed, over space and time from partial observations (Zhang et al. 2023). Reliable TSE support highways management, including variable speed limits, ramp metering, and incident detection, but in practice traffic data are sparse, noisy, and heterogeneous (Chen et al. 2001). Loop detectors (Eulerian data) and contemporary technologies e.g., a small fraction of connected vehicles or probes (Lagrangian data) provide only partial information of the traffic flow, often limited to speed at a few locations, while the full spatio-temporal distributions of speed and density remain unobserved (Duret and Yuan 2017).

Classical TSE methods are mainly model-driven. They combine macroscopic traffic flow models such as first-order Lighthill Whitham Richards (LWR)-type (Imran et al. 2021), and second-order Payne-Whitham-type (H. Khan et al. 2019), or Aw-Rascle-type (Hamori and Tan 2026) models with data assimilation schemes (e.g. Kalman filters) to estimate the hidden state from sparse sensor data. These methods exploit well-understood traffic physics, but they rely on simplified assumptions, fixed model parameters, and often first-order or simplified second-order dynamics that may not capture important non-equilibrium effects such as stop-and-go waves, hysteresis, or realistic wave propagation (Imran et al. 2023). Their performance can degrade when the real traffic deviates from the assumed model, or when only partial variables are observed (e.g., speeds without densities).

Data-driven approaches, especially deep learning (DL), offer the opposite trade-off. Pure neural-network models can approximate highly nonlinear input-output relationships directly from data, and have been applied to traffic prediction and TSE using detector measurements and probe trajectories (Polson and Sokolov 2018). However, these methods treat traffic as just another spatio-temporal signal. They require large, high-quality datasets (Shi et al. 2021), provide limited physical interpretability, and can produce non-physical estimates (e.g. negative densities, unrealistic wave speeds) when presented with sensor gaps or out-of-distribution conditions. Their generalization beyond the training regime is therefore questionable in safety-critical highway control.

Meanwhile, physics-informed deep learning (PIDL) has emerged as a hybrid paradigm that combines the previous two philosophies (Huang and Agarwal 2022). In PIDL, neural networks approximate the unknown traffic variables (e.g., speed and density), while the governing partial differential equations (PDEs) of traffic flow appear explicitly in the loss function as soft constraints (Huang et al. 2024). Residuals of the model are penalized at collocation points (considered as virtual detectors) throughout the domain, so that the learned fields are consistent with both the available data and the underlying dynamics. Recent studies have embedded first-order LWR (Huang and Agarwal 2020) or second-order Aw-Rascle-Zhang (ARZ) models (Shi et al. 2021) into PIDL architectures for TSE and parameter discovery, demonstrating improved data efficiency and accuracy compared to purely data-driven baselines.

Despite this progress, existing PIDL-based TSE methods still have limitations for real-world highway applications. First, most works require both density and velocity measurements as inputs, even though in many real applications only speeds (from loop detectors or probes) are reliably available, while densities are noisy, aggregated, or entirely missing. The design of the learning architecture and loss function is typically not tailored to the “velocity-only” scenarios, density is treated as an observed variable rather than a hidden field inferred from physics, hard physical bounds on density are rarely enforced, and behavior-related parameters are often fixed rather than learned jointly with the traffic state. Second, the traffic models used as constraints are often relatively weak or simplified (e.g. first-order LWR or specific ARZ formulations), and do not fully exploit richer second-order dynamics with explicit wave-propagation and relaxation dynamics. Thus, in the light of this rationale, the contributions of this paper are:

1. We propose a velocity-only PIDL framework for TSE, in which a second-order Jiang-Qing-Zhu (JQZ) model (Jiang et al. 2002) is used to couple speed and density, while only sparse speed data are required. Density is fully unobserved and inferred as a latent field consistent with the traffic PDEs.

2. We design a dedicated neural architecture and constraint strategy that enforces physically meaningful bounds on density, and jointly learns key behavioral parameters of the model together with the traffic state, instead of assuming them fixed or calibrated offline.

3. We design a loss function that combines data misfit, PDE residuals, initial conditions and parameter priors, and we systematically compare the resulting PIDL estimator with a DL-only baseline to quantify the benefits of the traffic model, the architecture and the loss design.

Together, these elements give a TSE method that is better aligned with real-world highway sensing conditions, exploits a more realistic second-order description of traffic dynamics, and leverages physics to constrain DL in regions where data are scarce or even absent.

The rest of this paper is organized as: section 2 presents the methods. Sections 3 and 4 are the case study and the results, while sections 5 and 6 present the discussion and conclusions of the study, respectively.

2. METHODS

This section describes the second–order macroscopic traffic model and the PIDL architecture used to reconstruct traffic states from sparse velocity measurements.

2.1 The traffic flow model

In this study, we consider a road segment with position $x \in [0, L]$ and time $t \in [0, T]$. Traffic is represented at the macroscopic level by density $\rho(x,t)$ and speed $v(x,t)$, the flow is given by $q(x,t) = \rho(x,t).v(x,t)$. Conservation of vehicles over any control volume gives the continuity equation, which is

$$[1] \quad \frac{\partial \rho}{\partial t}(x,t) + \frac{\partial}{\partial x}(\rho(x,t)\, v(x,t)) = 0,$$

it states that the local rate of change of density is balanced by the divergence of the flow (Hadadi and Aghayan 2025). The evolution of the speed is described by a second–order balance law of the form

$$[2] \quad \frac{\partial v}{\partial t}(x,t) + (v(x,t) - c_0)\, \frac{\partial v}{\partial x}(x,t) - \frac{V(\rho(x,t)) - v(x,t)}{\tau} = 0,$$

where c_0 is the backward propagation speed of perturbations (wave speed) and τ is a relaxation time. The convective term $(v(x,t) - c_0)\, v_x$ controls the propagation of speed perturbations along the road, while the relaxation term $(V(\rho(x,t)) - v(x,t))/\tau$ models the tendency of drivers to adapt their speed towards an equilibrium value depending on density (Khan et al. 2020). Whereas, the $V(\rho(x,t))$ (equilibrium speed-density relation) is specified by the Greenshields model that is

$$[3] \quad V(\rho(x,t)) = v_f(1 - \frac{\rho(x,t)}{\rho_m}),$$

where v_f is the free-flow speed and ρ_m is the jam density. For $\rho(x,t) = 0$ one obtains $V(0) = v_f$, while for $\rho(x,t) = \rho_m$ one has $V(\rho_m) = 0$, it results in a linear decrease of desired speed with increasing density. In the present study v_f is kept fixed, whereas the model parameters (c_0, τ, ρ_m) are estimated from data within the PIDL framework.

The employed second-order JQZ model (combined Eqs. 1 and 2) is better suited to our proposed architecture for several reasons compared with the ARZ formulation adopted in recent PIDL-based TSE studies (Shi et al. 2021). First, the velocity equation combines a convective term $(v - c_0)\, v_x$ with an explicit relaxation towards the equilibrium speed $V(\rho)$, so that the same speed-density relation governs both stationary and dynamic behaviour. This results in a direct and transparent link between model parameters (c_o, τ, ρ_m), which is particularly important when these parameters are learned from data. Second, the model evolves the physically meaningful pair (ρ, v) rather than the ARZ "generalized velocity" variable, which simplifies the coupling with sparse speed measurements and avoids ambiguities in interpreting the latent state. Third, the chosen dynamics have been shown to provide robust descriptions of stop-and-go waves and non-equilibrium transitions, while maintaining a clear equilibrium limit defined by the fundamental diagram. In the context of PIDL with velocity-only data, this structure offers a more interpretable and stable foundation compared to other models (e.g., the ARZ model), without introducing additional unobserved quantities that are weakly constrained by the available measurements.

2.2 The design of the PIDL architecture for TSE

This section presents the overall design of the PIDL architecture employed in this study.

2.2.1 Neural parameterization of speed and density

Let $\Omega = [0, L] \times [0, T]$ denote the space-time domain. The PIDL model represents the state variables $\rho(x,t)$ and $v(x,t)$ by two feed-forward neural networks. To improve conditioning, the inputs are first mapped to the square $[-1,1]^2$,

$$[4] \quad \tilde{x} = 2\, \frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1, \tilde{t} = 2\, \frac{t - t_{\min}}{t_{\max} - t_{\min}} - 1,$$

where $[x_{\min}, x_{\max}]$ and $[t_{\min}, t_{\max}]$ are the spatial and temporal bounds of the data grid. The speed is approximated by

[5] $\hat{v}(x,t) = \mathcal{N}_v(\tilde{x}, \tilde{t}; \theta_v),$

where $\mathcal{N}_v$ is a fully connected network with parameters θ_v. Density is not predicted directly but obtained from a second network $\mathcal{N}_\rho$ with parameters θ_ρ,

[6] $\hat{r}(x,t) = \mathcal{N}_\rho(\tilde{x}, \tilde{t}; \theta_\rho), \hat{\rho}(x,t) = \rho_m\, \sigma(\hat{r}(x,t)),$

where σ is the logistic sigmoid. The scaling by ρ_m enforces the physical bound $0 < \hat{\rho}(x,t) < \rho_m$ for all $(x,t) \in \Omega$. Throughout the paper, a common architecture is used for $\mathcal{N}_v$ and $\mathcal{N}_\rho$ (same number of layers and neurons with $\tanh$ activations), but the weights θ_v and θ_ρ are distinct. The physical parameters c_0 and τ are treated as trainable scalars, constrained to lie in prescribed intervals, and penalized around prior values c_0^{prior} and τ^{prior}. The jam density ρ_m can either be fixed a priori or learned in the same way (in this paper it is learned in the same way).

2.2.2 Data sets and collocation points

The study assumes access to sparse velocity measurements**,** and optionally, an initial speed profile, but no density observations. The set of observed speeds is denoted by

[7] $\mathcal{O} = \{(x^{(i)}, t^{(i)}, v^{(i)})\}_{i=1}^{N_O},$

where $\left(x^{(i)}, t^{(i)}\right)$ are space-time coordinates of sensors or probe vehicles, $v^{(i)}$ are the corresponding measured speeds, and N_O is the number of observation points. At the initial time t_0, a speed profile is assumed known on a spatial grid $\{x_k\}_{k=1}^{N_x}$ with N_x points,

[8] $v_{t_0}(x_k) = v^*(t_0, x_k), k = 1, \dots, N_x.$

This provides an anchor for the solution at $t = t_0$ but does not constrain density, v^* is the known initial speed value. To enforce the PDEs (Eqs. 1 and 2) over the full domain, a set of collocation points

[9] $\mathcal{C} = \{(x_c^{(j)}, t_c^{(j)})\}_{j=1}^{N_C} \subset \Omega,$

is generated independently of the sensor locations, where N_C denotes the number of collocation points. In practice, $\mathcal{C}$ is obtained by random sampling in Ω and augmented with the observation locations to ensure that physics is enforced both in-between and near the data points. At each $(x_c, t_c) \in \mathcal{C}$, the networks $\hat{v}$ and $\hat{\rho}$ and their derivatives with respect to t and x are evaluated by automatic differentiation.

The density scale at the initial time is weakly constrained by a small set of initial-condition points

[10] $\mathcal{I} = \{x_{\text{IC}}^{(\ell)}\}_{\ell=1}^{N_{\text{IC}}},$

with N_{IC} locations along $t = t_0$, where $\hat{\rho}(x_{\text{IC}}, t_0)$ is penalized against a light-traffic value ρ_{IC}, and ℓ represents points where the initial density conditions are imposed. This prevents trivial or degenerate density without imposing a fully specified initial profile.

2.2.3 Physics informed composite objective function

Training of the PIDL model is formulated as the minimization of a composite loss that combines data misfit, PDE residuals and regularization terms. All components are expressed as mean squared errors, and all symbols are defined when introduced.

Thus, the velocity data misfit quantifies the agreement between predicted and observed speeds at the points in $\mathcal{O}$:

$$[11] \quad J_{\text{data}} = \frac{1}{N_O} \sum_{(x^{(i)},t^{(i)})\in\mathcal{O}} (\hat{v}(x^{(i)},t^{(i)}) - v^{(i)})^2.$$

Similarly, the initial–time speed misfit enforces consistency with the known speed profile at $t = t_0$:

$$[12] \quad J_{t_0} = \frac{1}{N_x} \sum_{k=1}^{N_x} (\hat{v}(x_k,t_0) - v_{t_0}(x_k))^2.$$

The continuity equation Eq. 1 is imposed at collocation points via the residual

$$[13] \quad f_{\text{cont}}(x_c,t_c) = \hat{\rho}_t(x_c,t_c) + (\hat{\rho}(x_c,t_c)\,\hat{v}(x_c,t_c))_x,$$

and the corresponding loss

$$[14] \quad J_{\text{cont}} = \frac{1}{N_C} \sum_{(x_c,t_c)\in\mathcal{C}} f_{\text{cont}}\,(x_c,t_c)^2.$$

Here, $\hat{\rho}_t$ and $(\hat{\rho}\hat{v})_x$ denote partial derivatives with respect to t and x, evaluated by automatic differentiation. The velocity dynamics Eq. (2) are enforced analogously. At each collocation point,

$$[15] \quad f_{\text{vel}}(x_c,t_c) = \hat{v}_t(x_c,t_c) + (\hat{v}(x_c,t_c) - c_0)\,\hat{v}_x(x_c,t_c) - \frac{V(\hat{\rho}(x_c,t_c)) - \hat{v}(x_c,t_c)}{\tau},$$

and the associated loss is

$$[16] \quad J_{\text{vel}} = \frac{1}{N_C} \sum_{(x_c,t_c)\in\mathcal{C}} f_{\text{vel}}\,(x_c,t_c)^2.$$

The initial–density prior constrains the predicted density at the points in $\mathcal{I}$ towards ρ_{IC}:

$$[16] \quad J_{\text{IC}} = \frac{1}{N_{\text{IC}}} \sum_{x_{\text{IC}}\in\mathcal{I}} (\hat{\rho}(x_{\text{IC}},t_0) - \rho_{\text{IC}})^2,$$

where N_{IC} is the cardinality of $\mathcal{I}$. Finally, the parameter prior penalizes deviations of c_0, and $\boldsymbol{\tau}$ from prescribed prior values:

$$[17] \quad J_{\text{prior}} = (c_0 - c_0^{\text{prior}})^2 + (\tau - \tau^{\text{prior}})^2.$$

Collecting all trainable quantities in $\Theta = (\theta_v, \theta_\rho, c_0, \tau, \rho_m)$, the total physics–informed loss is defined as

$$[18] \quad W_{\text{data}}\,J_{\text{data}} + W_{t_0}\,J_{t_0} + W_{\text{cont}}\,J_{\text{cont}} + W_{\text{vel}}\,J_{\text{vel}} + W_{\text{IC}}\,J_{\text{IC}} + W_{\text{prior}}\,J_{\text{prior}},$$

where $W_{\text{data}}, W_{t_0}, W_{\text{cont}}, W_{\text{vel}}, W_{\text{IC}}, W_{\text{prior}} > 0$ are pre-specified weights controlling the relative importance of the different components.

2.2.4 Data driven baseline and training procedure

For comparison, a purely data–driven baseline is obtained by training only the speed network $\mathcal{N}_v$ on the observation data and initial profile, without a density network, PDE residuals or parameter learning. Its objective is

$$[19] \qquad J_{\text{DL}}(\theta_v) = W_{\text{data}} J_{\text{data}} + W_{t_0} J_{t_0},$$

with the same definitions of J_{data} and J_{t_0} as above. Both the PIDL and DL models are trained using the Adam optimizer. At each iteration, a mini–batch of collocation points is drawn from $\mathcal{C}$, the losses Eqs. 11 and 17 are evaluated on the current networks, and the gradients of J_{PIDL} (or J_{DL}) with respect to Θ are computed by automatic differentiation. The parameters are then updated in the direction of steepest descent. After convergence, the PIDL model gives reconstructed fields $\hat{v}(x,t)$ over Ω, together with estimates of the behavioral parameters (c_0, τ, ρ_m), using only sparse velocity measurements as direct supervision.

3. SIMULATION SETUP AND SCENARIOS

The simulation settings and test scenarios are described in this section. We use a dataset generated with the METANET (Papageorgiou et al., 1989) macroscopic traffic simulator for a 2000 m highway segment over a 600 s horizon as shown in Figure 1(a). It provides a reference speed on a regular space-time grid. The data set presents a challenging traffic scenario where two traffic shockwaves are resulted from road closure and red light, respectively. At 2000 m, the traffic is blocked for the first 300 s into the simulation. Next, a red light at 1000 m is activated for 50 s resulting in a small, yet strong shockwave between 150 s and 300 s. Sparse sensor velocities are extracted from this grid to form the observation set $\mathcal{O}$, while the full speed observations are retained only for a posteriori evaluation.

The proposed PIDL model couples a JQZ model with a Greenshields equilibrium relation, a velocity-only observation model, and two fully connected neural networks representing speed and latent density. Each network has 4 hidden layers with 40 neurons per layer and tanh activation. Physics constraints are enforced at a set of collocation points $\mathcal{C}$, obtained by random sampling over the space-time domain and augmented with all observation locations. A light prior on initial density is imposed at $N_{\text{IC}} = 300$ points along the initial time slice. Key model parameters, neural architectures, and training hyperparameters are summarized in Table 1.

To assess the robustness of the approach, we consider two scenarios:

- **Scenario 1 - Eulerian (fixed detectors):** Velocity observations are sampled at a finite number of fixed locations along the road (virtual loop detectors), over the full time horizon. This emulates a conventional highway instrumented with stationary sensors providing Eulerian measurements.

- **Scenario 2 - Lagrangian (probe vehicles):** A subset of vehicle trajectories is selected and treated as probe vehicles providing Lagrangian speed measurements. These vehicles are assumed to travel through the domain with a given penetration rate (PR) such as 15%, generating probe-like observations along their paths. No density measurements are used, only their velocity traces are provided to the PIDL model.

In both scenarios, the same underlying traffic model, neural architectures, and training settings are used, only the construction of the observation set $\mathcal{O}$ differs.

Table 1: Simulation parameters

Parameters	Symbol/value	Description
Free–flow speed	$v_f = 30.0$ m/s	Upper bound of equilibrium speed.
Jam density	$\rho_m \in [0.08, 0.20]$ veh/m	Jam density; density constrained to $0 < \rho < \rho_m$

Perturbation speed	$c_0 \in [5.0,50.0]$ m/s	Initial value $c_0^{(0)} = 15.0$ m/s.
Relaxation time	$\tau \in [3.0,40.0]$ s	Initial value $\tau^{(0)} = 15.0$ s.
Initial density prior	$\rho_{\text{IC}} = 0.03$ veh/m	Light initial density at $N_{\text{IC}} = 300$ points at $t = t_0$.
Observation set	$\mathcal{O}$, size N_O	Sparse velocity measurements $v^{(i)}$ at $(x^{(i)}, t^{(i)})$ from the grid.
Collocation pool	$\mathcal{C}$, size $N_C \approx N_F + N_O$	$N_F = 4{,}000$, plus all observation locations.
Velocity network	$\mathcal{N}_v$	Fully connected NN, 4 hidden layers, 40 neurons each, $\tanh$ activation; inputs scaled to $[-1,1]^2$.
Density network	$\mathcal{N}_\rho$	Same architecture as $\mathcal{N}_v$
DL training steps	2400	Adam iterations for the DL-only baseline.
PIDL training steps	2400	Adam iterations for the PIDL.
Loss weights	$W_{\text{data}}, W_{t_0}, W_{\text{cont}}, W_{\text{vel}}, W_{\text{IC}}, W_{\text{prior}}$	Set to $1.0, 0.5, 2.0, 2.0, 0.1, 10^{-4}$, respectively.

4. RESULTS

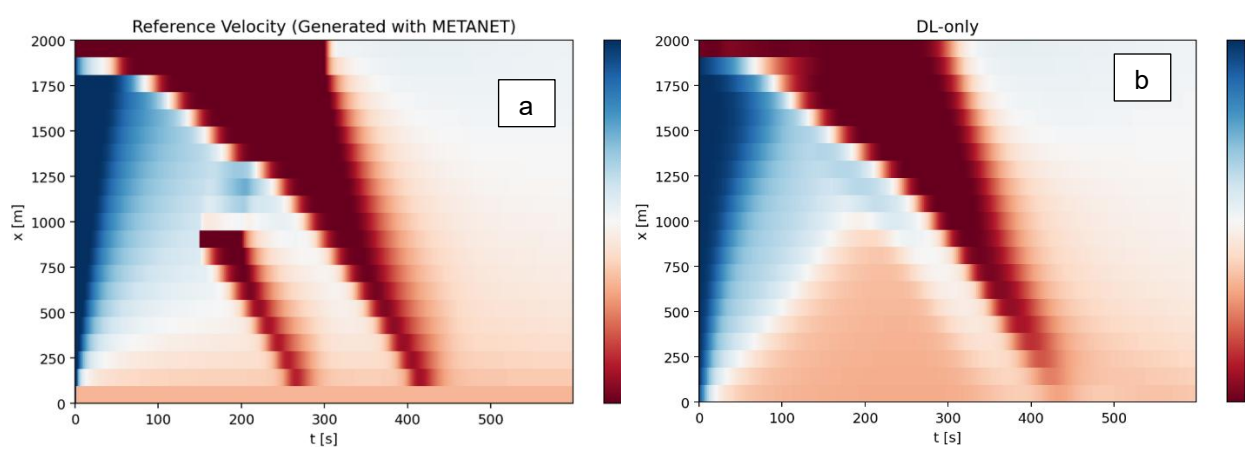
The results of this study are reported in this section for the two observation scenarios, and the corresponding reconstructions are shown in Figures 1 and 2. Figure 1(a) presents the baseline (reference) speed used as ground truth for evaluation in both scenarios,. Model accuracy is quantified using the root mean square error (RMSE), which is reported for both the DL benchmark and the proposed PIDL model.

4.1 Scenario 1 – Eulerian (fixed detectors):

In this scenario, the observation set $\mathcal{O}$ comprises velocity measurements collected at a finite number of fixed spatial locations (virtual loop detectors) over the full time horizon. The resulting reconstructions from the DL benchmark and the proposed PIDL model are shown in Figures 1(b) and 1(c), respectively.

Overall, the PIDL model provides superior performance in both qualitative fidelity and quantitative accuracy. The DL model captures the dominant shockwave associated with the downstream blockage at $x = 2000$ m, however, it fails to reproduce the secondary shockwave induced by the internal red-light at $x = 1000$ m, which is evident in the reference velocity in Figure 1(a) but largely missing in Figure 1(b). Consistent with this visual discrepancy, the DL model attains an RMSE of 3.35.

In contrast, the proposed PIDL model accurately reconstructs both shockwave patterns, including the secondary wave generated at $x = 1000$ m, as shown in Fig. 1(c). This improved reconstruction is reflected in a substantially lower RMSE of 1.98, confirming the advantage of physics regularization under sparse, fixed-location sensing.



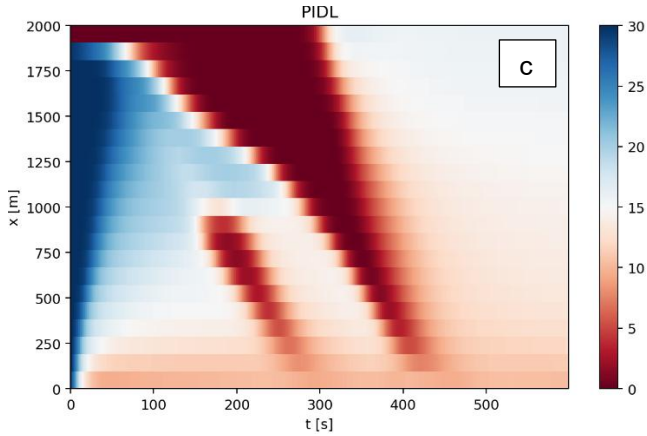


Figure 1: Evolution of traffic speed over space-time grid. (a) Reference speed evolution over the grid generated with METANET model, (b) reconstructed speed evolution by the DL model, and (c) the reconstructed speed evolution by the proposed PIDL model.

4.2 Scenario 2 - Lagrangian (probe vehicles):

In this scenario, the observation set $\mathcal{O}$ consists exclusively of Lagrangian (probe-like) velocity measurements. Specifically, a 15 % PR is assumed, and the corresponding trajectory samples are distributed over the space-time domain as illustrated in Figure 2(a) and 2(b). The DL benchmark reconstruction is reported in Figure 2(a). The DL model reproduces the reference velocity with good qualitative agreement, capturing both shockwave patterns observed in the ground truth. This visual consistency is supported by an RMSE of 2.14.

The PIDL reconstruction is shown in Figure 2(b). Relative to the DL benchmark, PIDL yields a closer match to the reference field, with sharper and more coherent shockwave structures, resulting in a lower RMSE of 1.52. Overall, PIDL consistently outperforms the purely data-driven baseline under Lagrangian sensing, indicating that embedding the traffic-flow physics improves reconstruction accuracy even when probe trajectories provide broader space-time coverage.

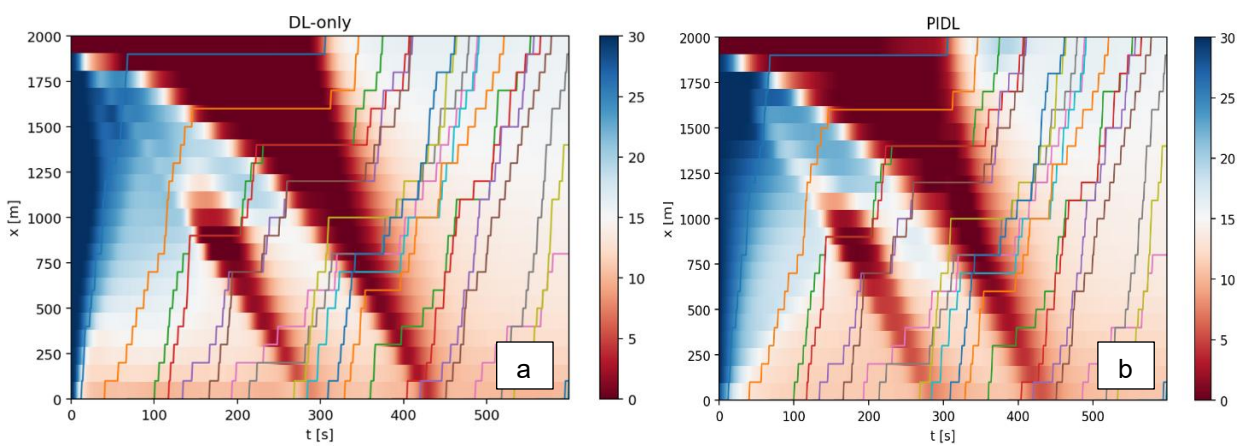


Figure 2: Evolution of traffic speed over space-time grid, (a) reconstructed speed evolution by the DL model, and (b) the reconstructed speed evolution by the proposed PIDL model.

5. DISCUSSION

This study investigates velocity-only TSE, where the objective is to reconstruct the space-time evolution of traffic speed from partial speed observations. This problem is aligned with real highway sensing, where

loop detectors provide Eulerian speed samples at a few fixed locations and probe data provide Lagrangian speed samples along a limited set of trajectories, while density is typically unobserved or unreliable. In such settings, reconstructing non-equilibrium dynamics (e.g., traffic shockwaves) is challenging because sparse velocity data alone do not uniquely determine the full space-time field. To address this gap, we proposed and evaluated a velocity-only PIDL model that embeds a second-order JQZ traffic model into the learning process, while using only sparse velocity observations for training. The estimator learns two continuous fields, $v(x,t)$ and a latent $\rho(x,t)$, and enforces the traffic PDE residuals at collocation points across the domain. We compared this approach to a DL-only baseline under two sensing configurations such as Eulerian fixed detectors and Lagrangian probe trajectories, using a METANET-generated reference speed for a posteriori evaluation.

The results show that PIDL provides more accurate and more dynamically consistent reconstructions than the DL baseline in both scenarios. In the Eulerian case (scenario 1), where observations are spatially sparse, the DL model captures the dominant downstream shockwave but fails to reproduce the secondary shockwave induced by the internal red-light, leading to higher error (RMSE 3.35). PIDL reconstructs both wave patterns and reduces the RMSE to 1.98. In the Lagrangian case (scenario 2), the DL model improves due to increased space-time coverage and captures both shockwaves (RMSE 2.14), yet PIDL remains more accurate (RMSE 1.52) and yields a closer match to the reference shockwave patterns.

These gains follow directly from the technical design of PIDL model. The DL-only baseline primarily relies on statistical interpolation/extrapolation from the observation set, therefore, it is more likely to miss or smooth shockwave features that are not directly constrained by measurements. By contrast, PIDL introduces a physics-consistency mechanism that is the JQZ residual constraints are enforced throughout the domain, which regularizes the reconstruction toward physically admissible wave propagation and enables information to be propagated beyond sensor locations through the governing dynamics. Moreover, treating density as a latent field, instead of requiring density measurements, leverages the coupling structure of the second-order model while remaining compatible with realistic velocity-only sensing. Together, these elements explain why PIDL better preserves shockwave patterns and achieves lower RMSE under both Eulerian and Lagrangian observation settings.

6. CONCLUSION

This study proposes velocity-only TSE, where the goal is to reconstruct the space-time speed from sparse speed observations under realistic sensing. We proposed a velocity-only PIDL model that embeds a second-order JQZ traffic model. The method learns continuous speed field using neural network, while enforcing the governing dynamics through model-residual penalties at collocation points across the space-time domain. On a METANET-generated benchmark, PIDL outperformed a DL-only baseline. For Eulerian fixed detectors, PIDL better reconstruct shockwave dynamics and reduced RMSE from 3.35 to 1.98. For Lagrangian probe trajectories (15% penetration), PIDL further improved accuracy (RMSE 1.52 vs. 2.14). These results indicate that embedding second-order physics improves identifiability and stabilizes reconstruction away from measurements, making the approach suitable for highway monitoring and control-oriented TSE (e.g., shockwave/queue tracking) using mixed detector-probe speed data.

6.1 Limitations and future work

The present evaluation is conducted on simulated macroscopic data, which offers a controlled benchmark but does not fully capture field-specific effects such as sensor noise, aggregation errors, detector dropouts, and behavioural heterogeneity. In addition, the Lagrangian setting is assessed using a single probe PR, and therefore the sensitivity of the estimator to different PRs and non-uniform (biased) trajectory coverage is not yet quantified. Future work will focus on validating the proposed PIDL framework on real-world datasets with realistic noise and missing-data, and on systematically investigating robustness across a wider range of probe penetration rates and sampling biases to better reflect operational deployments.

DATA AND CODES

The dataset used in this study and the Python implementation of the proposed PIDL model are made publicly available on GitHub at https://github.com/waheedemran/PIDL-model.

REFERENCES

Chen, Haibo, Susan Grant-Muller, Lorenzo Mussone, and Frank Montgomery. 2001. "A Study of Hybrid Neural Network Approaches and the Effects of Missing Data on Traffic Forecasting." *Neural Computing & Applications* 10 (3): 277–86. https://doi.org/10.1007/s521-001-8054-3.

Duret, Aurélien, and Yufei Yuan. 2017. "Traffic State Estimation Based on Eulerian and Lagrangian Observations in a Mesoscopic Modeling Framework." *Transportation Research Part B: Methodological* 101 (July): 51–71. https://doi.org/10.1016/j.trb.2017.02.008.

H. Khan, Zawar, Waheed Imran, Sajid Azeem, Khurram S. Khattak, T. Aaron Gulliver, and Muhammad Sagheer Aslam. 2019. "A Macroscopic Traffic Model Based on Driver Reaction and Traffic Stimuli." *Applied Sciences* 9 (14): 2848. https://doi.org/10.3390/app9142848.

Hadadi, Farhad, and Iman Aghayan. 2025. "An Extended LWR Non-Lane-Based-Discipline Model for Light Vehicle Traffic Flow Analysis Incorporating Driver Behavior." *Chinese Journal of Physics* 97 (October): 591–606. https://doi.org/10.1016/j.cjph.2025.06.049.

Hamori, Thomas, and Changhui Tan. 2026. "On the Aw–Rascle–Zhang Traffic Models with Nonlocal Look-Ahead Interactions." *Nonlinear Analysis* 262 (January): 113930. https://doi.org/10.1016/j.na.2025.113930.

Huang, Archie J., and Shaurya Agarwal. 2020. "Physics Informed Deep Learning for Traffic State Estimation." *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, September 20, 1–6. https://doi.org/10.1109/ITSC45102.2020.9294236.

Huang, Archie J., and Shaurya Agarwal. 2022. "Physics-Informed Deep Learning for Traffic State Estimation: Illustrations With LWR and CTM Models." *IEEE Open Journal of Intelligent Transportation Systems* 3: 503–18. https://doi.org/10.1109/OJITS.2022.3182925.

Huang, Archie J., Animesh Biswas, and Shaurya Agarwal. 2024. "Incorporating Nonlocal Traffic Flow Model in Physics-Informed Neural Networks." *IEEE Transactions on Intelligent Transportation Systems* 25 (11): 16249–58. https://doi.org/10.1109/TITS.2024.3429029.

Imran, Waheed, Zawar H. Khan, T. A. Gulliver, Khurram S. Khattak, Salman Saeed, and M. Sagheer Aslam. 2021. "Macroscopic Traffic Flow Characterization for Stimuli Based on Driver Reaction." *Civil Engineering Journal* 7 (1): 1–13. https://doi.org/10.28991/cej-2021-03091632.

Imran, Waheed, Zawar H. Khan, T. Aaron Gulliver, Muhammad Alam, and Khurram S. Khattak. 2023. "Non-Homogeneous Traffic Characterization Based on Driver Reaction and Stimuli." *Transportation Research Interdisciplinary Perspectives* 21 (September): 100858. https://doi.org/10.1016/j.trip.2023.100858.

Jiang, Rui, Qing-Song Wu, and Zuo-Jin Zhu. 2002. "A New Continuum Model for Traffic Flow and Numerical Tests." *Transportation Research Part B: Methodological* 36 (5): 405–19. https://doi.org/10.1016/S0191-2615(01)00010-8.

Khan, Zawar H., Waheed Imran, Thomas Aaron Gulliver, Khurram S. Khattak, Zahid Wadud, and Akhtar Nawaz Khan. 2020. "An Anisotropic Traffic Model Based on Driver Interaction." *IEEE Access* 8: 66799–812. https://doi.org/10.1109/ACCESS.2020.2985668.

Papageorgiou, Markos, Jean-Marc Blosseville, and Habib Hadj-Salem. "Macroscopic modelling of traffic flow on the Boulevard Périphérique in Paris." *Transportation Research Part B: Methodological* 23, no. 1 (1989): 29-47.

Polson, Nicholas, and Vadim Sokolov. 2018. "Bayesian Particle Tracking of Traffic Flows." *IEEE Transactions on Intelligent Transportation Systems* 19 (2): 345–56. https://doi.org/10.1109/TITS.2017.2650947.

Shi, Rongye, Zhaobin Mo, and Xuan Di. 2021. "Physics-Informed Deep Learning for Traffic State Estimation: A Hybrid Paradigm Informed By Second-Order Traffic Models." *Proceedings of the AAAI Conference on Artificial Intelligence* 35 (1): 540–47. https://doi.org/10.1609/aaai.v35i1.16132.

Zhang, Zhao, Xianfeng Terry Yang, and Hao Yang. 2023. "A Review of Hybrid Physics-Based Machine Learning Approaches in Traffic State Estimation." *Intelligent Transportation Infrastructure* 2 (May): liad002. https://doi.org/10.1093/iti/liad002.